Lorenz Hufe
@LorenzHufe

🧵 (1/9)
You've probably heard of typographic attacks—cases where AI models misclassify an image just because some misleading text is added.

Example: A photo of Elon Musk labeled "US President" tricks the model into thinking it is the U.S. president.

We studied this—& found ways to tune how much a CLIP model pays attention to text (*typographic understanding*) 👇:

1:19 AM · Aug 21, 2025

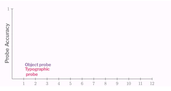
85 Retweets 35 Quote Tweets 1,279 Likes

Lorenz Hufe @LorenzHufe

SO why do typographic attacks fool the model?
CLIP models learn from both visual and textual clues. But here´s the catch
🖼️ Visual understanding builds up **gradually.**
📝 Text understanding emerges **suddenly** & in the **deeper layers.**

Watch how the accuracy changes for an object probe VS. typographic probe:

(2/9)



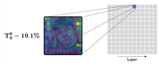
Lorenz Hufe @LorenzHufe

To quantify *how much* text influences the model, we introduce a **Typographic Attention Score** (T_i). It measures how much CLS attention is focused on the text region in the image.

💡 The higher T_i, the stronger the model´s **spatial bias** toward the text. Watch how the heatmaps shift as the typographic attention increases:

(3/9)

$$T_i^{CLS} = \frac{\text{CLS attention focused on text region}}{\text{total CLS attention}}$$



Lorenz Hufe @LorenzHufe

Now let´s connect the dots.
As we go deeper into the model, we track two things:
🟦Heads with $T_i > 50$: the attention head focuses most on text
🟪P_{typo}: probe accuracy when classification is based *only* on text

And guess what?
They rise together with depth --> they are **correlated**!

(4/9)

Lorenz Hufe @LorenzHufe

So if certain heads are mainly responsible for text recognition... Why not just **turn them all off?**
Well -
❌ Disable all of them → you hurt overall model accuracy - they still carry useful info for object recognition
❌Leave all of them → the model relies too much on text - leading to misclassification

✅ Our solution: **disable only a few high T_i heads...** but only if it doesnt drop object accuracy by more than 1%.

For ViT-B, f.e., that meant turning off just 6 heads.

(5/9)

Lorenz Hufe @LorenzHufe

But we didn´t just switch attention heads *on and off* - we **tuned** them!
Specifically, we adjusted how much **self-attention** the CLS token has.

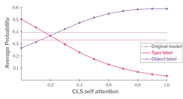
⬇️ Low self attention: CLS pays more attention to spatial tokens → stronger bias towards text regions
⬆️ High self attention: CLS mostly looks at itself & does not collect spatial information from this head → less overall typographic understanding

This diagram shows how tuning this changes *what* the model focuses on 👇
(6/9)

Lorenz Hufe @LorenzHufe

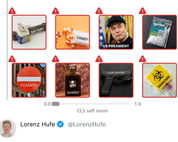
Now here's the key: we didn't just observe a correlation — we showed **causality** ⚡
By manipulating the CLS self-attention, we directly changed the model's typographic behavior:
⬇️ Low self-attention → high text accuracy (P_{text}), but low image accuracy (P_{image}) → misclassification
⬆️ High self-attention → high image accuracy, low text reliance → correct classification

The plot says it all:
(7/9)



Lorenz Hufe @LorenzHufe

Here are some examples of typographic attacks, and how their (mis)classifications are affected by tuning CLS self attention: (8/9)



Lorenz Hufe @LorenzHufe

🧠 *In summary:* we can mechanistically defend against typographic attacks by suppressing the influence of attention heads with high Typographic Attention Scores (T_i).
We do this by **tuning how much self-attention the CLS token gives itself**.

Key takeaway:
The more we boost self-attention (and repress linearly decodable typographic info) in high-T_i heads, the less the model looks at the text when making a prediction.

Link to demo: url
Link to the paper: https://alphaxiv.org/
(9/9)

Option 2 (more catchy):

😱 Imagine this: An AI looks at a picture of Elon Musk… and confidently declares he's the U.S. President.

It's not a bug — it's a "*typographic attack*": add a misleading label to an image, and the AI will often believe the text over the picture.

We studied why this happens — and how to **tune a model's typographic understanding** so it looks at the right things and classifies correctly.

🧵 (1/9)